

Lab 4: VibeCoding — Chat History

Repo: 43_reaction-time-tester | Date: 2026-09-30 | Tool: OpenCode (Muse Spark 1.3)

Objective: Use a Vibe Coding tool to fix broken code and add README features, in 3–4 prompt attempts. Clone repo, record 10s before video, fix via prompts, record 10s after video, push to personal repo (no PR to SETAPESU26). Deliverables under Lab-4 folder: videos, updated code, chat history.

1. Conversation log (prompts and outcomes)

#	User prompt	AI action / result
1	cd into project (shell)	Inspected README.md, main.py, game/game_engine.py, game/round.py, requirements.txt. Summarized Task 1–4 bugs.
2	“go through the readme, and fix all the tasks”	Fixed all 4 tasks: timing, game-over screen, replay+difficulties, sounds. Updated round.py, game_engine.py, main.py. Verified headless.
3	python main.py → command not found: python	Diagnosed: use python3 / venv python on macOS.
4	python3 main.py → No module named pygame	Diagnosed: system python has no pygame; must use .venv.
5	source .venv/bin/activate; python3 main.py → still No module named pygame	Diagnosed: venv activation issue in non-interactive shell; must use .venv/bin/python.
6	python main.py (in .venv) → pygame.font circular ImportError	Root cause: pygame 2.6.1 broken on Python 3.14.7 (font.py ↔ sysfont.py). Replaced with pygame-ce 2.5.8; updated requirements.txt.
7	“give this chats summary.. this is our assignment” + Lab-4 PDF brief	Provided chat summary for submission.
8	“videos folder created with before/after; generate pdf of the chat in this directory”	Generated this PDF (Lab4_Chat_History.pdf).

2. Code changes

game/round.py — register_input(): waiting → false_start (return None, no time); go → result (return now – go_time); result/false_start repeats ignored. Added state docs.

game/game_engine.py — handle_event() only acts in waiting/go, appends only valid times, plays false-start sound; update() plays go cue on waiting→go, pauses 800 ms (result) / 1000 ms (false-start), retries same round on false-start; _start_next_round() sets game_over + end jingle; reset(difficulty) with DIFFICULTIES Easy 1500–3500/3, Medium 1000–3000/5, Hard 500–2000/7; render() adds RED false-start + DARK game-over screen (all times, average, replay hints); sounds synthesized via array+math matching mixer format with graceful fallback.

main.py — loop honors engine.should_quit (Q/Esc on results screen).

requirements.txt — pygame → pygame-ce (2.5.8) for Python 3.14 compat.

3. Verification

Headless (SDL dummy, pygame-ce): valid reaction ~123 ms (not inflated by ~2500 ms wait); false-start flagged, not recorded, same-round retry; 2-round session reaches game_over; game-over render ok; 1/2/3 replay to Easy/Medium/Hard; Q sets should_quit; go/false/end sounds created. Post env-fix: GameEngine

init+render+sounds ok in project .venv.

4. Environment fix

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pygame 2.6.1 + Python 3.14.7 fails at import (pygame/font.py line 17 ↔  
pygame/sysfont.py line 27 circular). Fix applied: .venv/bin/pip uninstall -y pygame;  
.venv/bin/pip install pygame-ce. requirements.txt now contains pygame-ce.
```

5. Deliverables checklist

videos/before.mov and videos/after.mov — present. Updated code — round.py, game_engine.py, main.py, requirements.txt. This PDF — Lab4_Chat_History.pdf. Push project to personal repo; do not PR SETAPESU26.